

Hao Liu

KAUST, Thuwal, Saudi Arabia

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Personal Profile

Hao Liu is a Ph.D candidate (**expected graduation: November 2026**) in the Accelerated Connected Computing Lab (ACCL, <https://cemse.kaust.edu.sa/acc1>) at KAUST. His research primarily focuses on enabling efficient and scalable DNN inference by leveraging the collaborative capabilities of edge and near-edge computing nodes. His research interests include distributed DNN inference, model compression and acceleration, and reconfigurable computing systems.

He received his Bachelor's degree in Electronic and Information Engineering and his Master's degree in Cyber Science and Technology from Beihang University, Beijing, China in 2019 and 2021, respectively.

Education

King Abdullah University of Science and Technology (KAUST)

Ph.D student in Computer Science (Advisor: Prof. Suhaib A. Fahmy)

Thuwal, Saudi Arabia

Jan 2022 - Present

Beihang University (BUAA)

Master of Cyber Science and Technology (Advisor: Prof. Zhenyu Guan)

Beijing, China

Sept 2019 - Jan 2022

Beihang University (BUAA)

Bachelor of Electronic and Information Engineering

Beijing, China

Sept 2015 - Jun 2019

Publications

Ask the Expert: Collaborative Inference for Vision Transformers with Near-Edge Accelerators

Hao Liu, Suhaib A. Fahmy

under review, 2026

Practical Modeling for Split DNN Inference on Near-Edge Accelerators

Hao Liu, Mohammed E. Fouda, Ahmed M. Eltawil, Suhaib A. Fahmy

IEEE International Conference on Edge Computing and Communications (IEEE EDGE), 2026

Split DNN Inference for Exploiting Near-Edge Accelerators

Hao Liu, Mohammed E. Fouda, Ahmed M. Eltawil, Suhaib A. Fahmy

IEEE International Conference on Edge Computing and Communications (IEEE EDGE), 2024

A Filter Rank Based Pruning Method for Convolutional Neural Networks

Hao Liu, Zhenyu Guan, Peng Lei

IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2021

A Wearable Ad Hoc Device for Situational Awareness and Trusted Collaboration

Zhenyu Guan, Jiawei Li, **Hao Liu**, Dawei Li

Smart Blockchain: Second International Conference, 2019

Physical unclonable functions for IoT device authentication

Zhenyu Guan, **Hao Liu**, Yuyao Qin

Journal of Communications and Information Networks (2019). 2019

Research Experience

LLM speculative decoding across edge and near-edge accelerators

Jeddah, Saudi Arabia

KAUST (Advisor: Prof. Suhaib A. Fahmy)

Feb 2026 - Now

- Investigate the predictability of token-level generation difficulty in LLM speculative decoding, and explored lightweight difficulty-aware mechanisms to guide decoding behavior.
- Design a collaborative generation scheme that reduces edge-cloud communication overhead and end-to-end latency without compromising generation quality based on the asymmetry between edge and cloud in compute, latency, and bandwidth.

Collaborative Inference for Vision Transformers with Near-Edge Accelerators

Jeddah, Saudi Arabia

KAUST (Advisor: Prof. Suhaib A. Fahmy)

Jun 2025 - Jan 2026

- Propose a novel collaborative inference framework for ViTs with a general formulation, leveraging a lightweight edge model and multiple near-edge experts to enhance inference efficiency and accuracy.
- Design a robust routing mechanism that exploits the Top- k predictions of the edge model to determine expert activation without overhead.
- Propose a progressive specialist training strategy, which explicitly enforces expert specialization while preserving general accuracy.
- Experiments on CIFAR-100 and ImageNet dataset using a real-world edge and near-edge testbed.

Practical Modeling for Split DNN Inference on Near-Edge Accelerators

Jeddah, Saudi Arabia

KAUST (Advisor: Prof. Suhaib A. Fahmy)

Jun 2024 - May 2025

- Show experimentally that autoencoder-based communication compression offers higher accuracy and compression ratio than quantization and tensor low-rank decomposition.
- Propose a precise computation and communication formulation that matches measurements from practical deployments, considering system characteristics of GPU data ingestion and packet-based network data transfer.
- Build a real system comprising multiple accelerators and validate our proposed split DNN approach with the VGG16 and ResNet50 models on the CIFAR-100 dataset in a single-split setup with different batch sizes.
- Show use of the performance model to guide multi-split, dynamic bandwidth, and multi-tenant search scenarios.

Network Architecture Search for Split Computing

Tsinghua University Internship (Advisor: Prof. Yu Wang)

Beijing, China

Jun 2023 - Sept 2023

- Learn and extend an existing project of Network Architecture Search (NAS) `aw_nas` (https://github.com/walkerning/aw_nas)

Split Computing for Deep Neural Networks

Jeddah, Saudi Arabia

KAUST (Advisor: Prof. Suhaib A. Fahmy)

Oct 2022 - May 2024

- Formulate the problem of multi-split on DNNs into an optimization problem considering the accuracy, computing and transmission overhead.
- Propose to insert multiple autoencoders to split DNNs and execute different partitions of DNNs on different devices to reach the trade-off between latency and energy.
- Experiment on ResNet50 and VGG16 with CIFAR100 and ImageNet dataset.

A Filter Rank Based Pruning Method for Convolutional Neural Networks

Beijing, China

Beihang University (Advisor: Prof. Zhenyu Guan)

Jun 2020 - Dec 2021

Physical Unclonable Functions for IoT Device Authentication

Beijing, China

Beihang University (Advisor: Prof. Zhenyu Guan)

Mar 2019 - Dec 2019

A Wearable Ad Hoc Device for Situational Awareness and Trusted Collaboration

Beijing, China

Beihang University (Advisor: Prof. Zhenyu Guan)

Sept 2018 - Feb 2019

Teaching Experience

Saudi Ministry of Interior Program

Riyadh, Saudi Arabia

Teaching Assistant for **ECE 265P: AI Training**.

April 2025, April 2026

KAUST

Jeddah, Saudi Arabia

Teaching Assistant for **CS256: Digital Design and Computer Architecture**

Sept 2022

Beihang University

Beijing, China

Teaching Assistant for the **Digital Circuits and Systems** course.

Sept 2020

Technical Projects

NNStreamer Deployment & Custom Plugin Development [Docker] [Code]

- Deployed the NNStreamer framework on NVIDIA Jetson edge platforms. Developed custom GStreamer plugins using C/C++ to optimize DNN collaborative inference across multiple devices.

ResNet Acceleration: Vitis-AI vs. hls4ml [Code]

- Implemented ResNet inference using the Xilinx Vitis-AI flow on Xilinx Alveo U55C. Conducted a comparative analysis against hls4ml implementations, evaluating performance metrics and resource utilization.

FPGA Game Design: Block the Brick [Code]

- Designed and implemented a "Block the Brick" game using Verilog HDL.

Skills

Programming High Performance Computing (HPC), Python (PyTorch), vllm, C/C++, Vitis-AI, Verilog, Jetson

English IELTS Overall Band 7 (Academic, L7.5 R8.5 W6.5 S6.0; taken on 7/31/2021)

Achievements

2019, 2020 **Second class**, The Graduate Academic Scholarships

Beijing, China

2018 **Second Prize**, RMB 30,000 funding supported by Student Research Training Program

Beijing, China

2018 **Second prize**, The 11th National College Student Information Security Contest (Captain)

Wuhan, China

2018 **Second prize**, The 28th Feng Ru Cup Student Academic Competition of Beihang University

Beijing, China

2017 **Third prize**, The 8th Lan Qiao Cup in MCU Design and Development Competition

Beijing, China